TASK SPECIFICATION: P1-04 PHASE P1 — FOUNDATIONS & DEVOPS INFRASTRUCTURE

PostgreSQL Persistence, Flyway Migrations & Testcontainers Harness

Git Branch: `feature/P1-04-database-flyway` · Target Pull Request: PR (feature/P1-04-database-flyway)

Core Objective & Technical Scope:
Establish PostgreSQL database connectivity, Flyway migration versioning (V1 baseline), and abstract Testcontainers integration test harness for real database testing.

Priority Level	High	Complexity Rating	`M`
Execution Status	Done ✅	Target PR / Commit	`PR (feature/P1-04-database-flyway)`
Upstream Dependencies	P1-03		
Architecture Blockers	None (Ready for execution)		

IMPLEMENTATION ROADMAP & ENGINEERING STEPS

1. Add Spring Data JPA, PostgreSQL JDBC driver, and Flyway migration dependencies.
2. Configure HikariCP connection pool with environment variable overrides.
3. Author V1__baseline.sql migration enabling pgcrypto and btree_gist extensions.
4. Configure Hibernate with ddl-auto=validate to enforce Flyway authority.
5. Create AbstractIntegrationTest spinning up reusable Testcontainers PostgreSQL instance.

TARGET FILE INVENTORY & EXPECTED ARTIFACTS

- `backend/pom.xml`
- `backend/src/main/resources/db/migration/V1__baseline.sql`
- `application.yml`
- `AbstractIntegrationTest.java`

ACCEPTANCE CRITERIA & QUALITY CHECKLIST

- Flyway migrations execute cleanly on clean database.
- Subsequent boots validate existing schema without schema drift.
- Integration tests run against real containerized PostgreSQL.

VERIFICATION & TEST COVERAGE STRATEGY

- Flyway baseline migration test
- Testcontainers repository integration tests